

Executive Summary

Across the United States, schools that face persistent funding challenges force teachers to personally finance classroom supplies or seek support through crowdfunding platforms like DonorsChoose. While DonorsChoose enables educators to request funding for materials ranging from basic classroom supplies to lab equipment and field trips, about one-third of posted projects go unfunded. Failure to reach these goals disproportionately affects students in under-resourced schools and limits their access to quality learning experiences. Our project aims to support DonorsChoose in selecting projects for review. Specifically, we built a predictive model to identify the 10% of newly posted projects most at risk of going unfunded without timely and personalized intervention. By focusing on the projects with the lowest predicted funding success, we aim to maximize the impact of human intervention.

Our analysis identified XGBoost as the most effective classification algorithm after evaluating several models using 5-fold stratified cross-validation (mean ROC AUC: 0.7663). Hyperparameter tuning via Grid Search further optimized the XGBoost model, achieving a best cross-validation ROC AUC of 0.7559. On the held-out test data, the final tuned model yielded an ROC AUC of 0.7572. To best align with the goal of identifying at-risk (unfunded) projects while managing intervention resources, we optimized the classification threshold using Youden's J statistic. This resulted in an optimal threshold of 0.68, at which the model achieves a Specificity of 0.68 (correctly identifying 68% of unfunded projects) and a Recall (Sensitivity) of 0.68 (identifying 68% of funded projects). This threshold provides a strong balance for targeting the bottom 10% of projects most likely to fail for review by the digital content expert.

In the future, it would be helpful to integrate our selected machine learning model with a database resource management system and a user interface for ad hoc analysis. This would result in a user-friendly decision support system (DSS). Adding intervention analysis would increase the efficiency of the intervention process and improve the operability of the model. However, our model did not include intervention analysis since we mainly based our recommended project list on invariable features (i.e., school information, project information).

Background and Introduction

The lack of proper funding for public schools across the United States is a major issue that continues to cause serious problems for students, teachers, and school communities. These problems go beyond just limited budgets: they create real inequalities between districts, force teachers to cover costs out of their own pockets, and push schools to rely on outside help like crowdfunding websites just to meet basic classroom needs. Even though there are efforts to close these gaps, a large number of school projects still don't get funded, which shows that deeper issues in the system remain unresolved.

In the U.S., public schools get money from three main sources: local governments, state governments, and the federal government. But nearly half of that money comes from local property taxes (Reschovsky, n.d.). That might seem fine at first, but it creates a huge problem. Areas with higher property values bring in more money for their schools. Meanwhile, schools in lower-income areas just don't have that kind of funding, even though they often serve students who need more support. This setup leads to big differences in the quality of education kids receive, depending largely on where they live (Farrie, Kim, 2024).

Some states do try to even things out by giving more funding to poorer districts through state aid. But according to research, these efforts do not always make up for the funding gaps caused by local property taxes (La Follette School of Public Affairs, 2023). The result is that many schools in low-income areas still can't offer the same opportunities, resources, or even safe facilities that wealthier districts take for granted. It's not only unfair but creates long-term disadvantages for the students who need help the most.

Because of these funding problems, teachers are often the ones who have to step up and make up the difference. It's quite common for teachers to buy supplies for their classrooms using their own money. In fact, it is reported that 90% of teachers in the U.S. spend out of pocket for things like pencils, paper, books, and even snacks or hygiene products for their students (Litvinov, 2022). On average, teachers spend more than \$500 a year, but in schools with higher poverty levels, it's often much more (Litvinov, 2022).

This situation puts a ton of pressure on teachers. Many are already dealing with large class sizes, limited support staff, and high expectations, and now they also have to worry about whether their students have the basics they need to learn. The stress of constantly having to fill in the gaps, both financially and emotionally, can lead to burnout. It's one of the reasons many teachers leave the profession, especially in underfunded districts (Peck, 2025).

Because school budgets often fall short, teachers and schools are turning to crowdfunding to try and get the supplies and support they need. Websites like DonorsChoose have become a popular platform for teachers to raise money for classroom projects. On DonorsChoose, teachers can post requests for specific items or experiences, like new books, STEM materials, or field trip funding, and donors from all over the country can choose to support them.

This system has helped a lot of classrooms, but it also has its limitations. According to Education Week, about one-third of the projects on DonorsChoose go completely unfunded (Superville, 2020). One reason is that smaller, cheaper projects (usually under \$500) are more likely to get funded (DonorsChoose). This, in turn, leads to the issue of equity. Teachers in more affluent

areas might have better access to social networks or donors who can help fund their projects, while teachers in high-need areas might struggle to get attention, even though their needs are just as urgent. So while crowdfunding has helped fill in some gaps, there is a need to understand how it can be implemented more effectively and efficiently.

Improving the funding success rate on DonorsChoose is critical because it directly enhances educational equity and student outcomes. When more projects are funded, more students gain access to essential resources that support their academic development. Our project aimed to support DonorsChoose in allocating its limited resources. Specifically, we built a predictive model to identify the 10% of newly posted projects most at risk of going unfunded. This model will inform the efforts of a digital content expert tasked with reviewing and improving select project descriptions to increase their likelihood of success. Once the digital content expert has reviewed projects, they will collaborate with the teachers to improve the postings. By focusing on the projects with the lowest predicted funding success, we aim to maximize the impact of human intervention. Our policy goals emphasize:

- **Efficiency:** Leveraging data to allocate limited editorial resources for maximum impact.
- **Equity:** Prioritizing support for projects in communities with the greatest financial need.
- **Effectiveness:** Improving overall project funding rates and user experience.

Related Work and the Status Quo Model

Both DonorsChoose and independent researchers have been working to figure out how to improve the success rates for the projects.

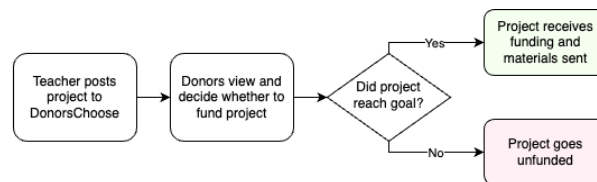


Figure 1. Flowchart diagram for status quo model.

Historically, projects relied solely on the strength of their original descriptions and visibility to attract donor support (Figure 1). There was no targeted intervention to improve the chances of underperforming projects. However, universal interventions were attempted, with DonorsChoose itself trying to help teachers increase their chances by offering suggestions and strategies through blog posts (DonorsChoose, 2023). These suggestions include keeping project costs under \$500, writing clear and engaging descriptions, sharing on social media, and making a strong case for how the materials will benefit students (DonorsChoose, 2023). While helpful, this advice provides very limited solutions, especially for teachers who need more expensive resources or who are less experienced in writing fundraising appeals.

Furthermore, some researchers have taken a data-driven approach. In one study, Ahmed et al. (2016) used machine learning to predict whether or not a project on DonorsChoose would get funded. They trained models using project data like cost, subject area, and school location to identify patterns in successful campaigns. The goal was to flag which projects were at risk of going unfunded so that teachers or the platform could take steps to improve their chances. This

study found a similar result that low-cost projects are most likely to be funded, and the location of the schools also played a big role.

Our project builds on this foundation, but it was primarily designed as a learning exercise for us to explore the end-to-end machine learning pipeline. While our goals were more exploratory, we also experimented with additional features and tested different models to see how they performed. Rather than aiming to outperform prior research, our focus was on understanding the process of building and evaluating predictive models in a real-world education context.

Problem Formulation and the Proposed Revised Model

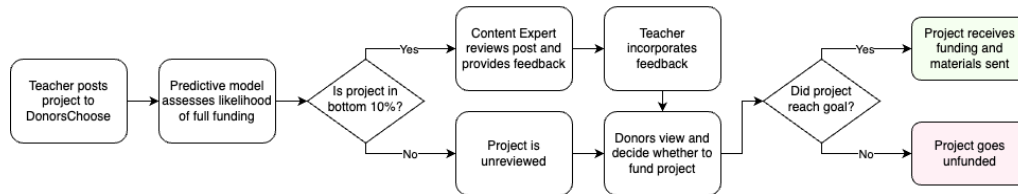


Figure 2. Flowchart diagram for revised model.

DonorsChoose recently introduced a human-centered intervention by hiring a digital content expert who reviews projects, helps teachers improve their postings, and increases their chances of reaching their funding threshold. However, this resource can only cover a small fraction of daily submissions. Given the expert’s limited capacity, the challenge is to determine which 10% of projects posted on a given day should be reviewed and supported. Our proposed data-driven model will optimize this intervention by flagging the projects most in need of support, i.e. least likely to receive full funding, early in the process (Figure 2). By identifying and recommending such projects early, we aim to improve funding outcomes, better the user experience, and help more students receive the materials they need to learn.

Data Availability and Exploratory Data Analysis

The dataset includes six tables, each providing different aspects of project funding. The names of these tables, along with a list of their columns, can be found in Supplementary Table 1. Each table is connected by the ‘projectid’ attribute. Based on the tables and their attributes, we mainly used the projects table and the “fully_funded” feature in the outcomes table.

Descriptive statistics were taken for each table, but only a short snippet will be shown below to conserve space (Figure 3). For the outcomes table, there were 619,326 unique ‘projectid’ values, all of which were given a “t/f” value for the ‘fully_funded’ status. Based on the statistics, it appears that we will need to clean the data, including converting some features into boolean values. Additionally, we noted that there are several missing values for the projects table, which may impact feature selection and data imputation (Supplementary Figure 1).

Outcomes Descriptive Statistics:				at_least_1_teacher_referred_donor fully_funded			
			projectid is_exciting				
count		619326	619326	count	524928	619326	
unique		619326	2	unique	2	2	
				top	f	t	
Data Types:				Missing Values:			
projectid			object	projectid		0	
is_exciting			object	is_exciting		0	
at_least_1_teacher_referred_donor			object	at_least_1_teacher_referred_donor		94398	
fully_funded			object	fully_funded		0	

Figure 3. Descriptive statistics snippet from outcomes table.

The projects table contains 664,098 projects submitted by 249,555 teachers. The higher number of projects than outcomes may be due to some projects not yet expired, as confirmed by annual funding outcomes (Figure 4). No projects from 2014 had outcomes, suggesting annual decisions.

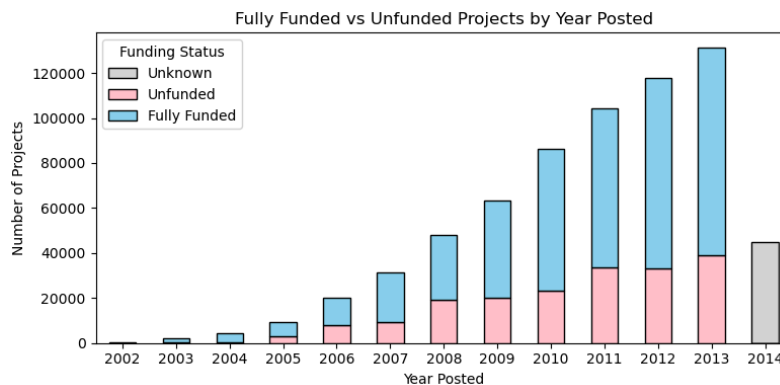


Figure 4. Funding status by year posted.

As shown in Figure 5, approximately one-third of the projects were not fully funded. This imbalance in funding outcomes will be taken into account when building and evaluating our predictive models. Additionally, we observe that only a small fraction of projects were labeled as “exciting,” suggesting that this feature may have limited coverage or strict criteria.

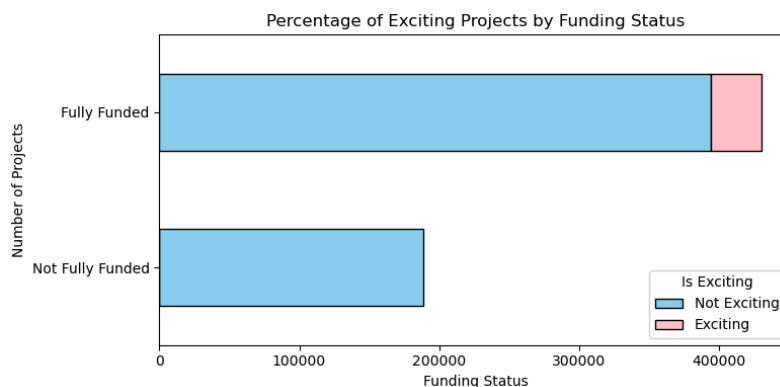


Figure 5. Percentage of exciting projects by funding status.

As part of our exploratory data analysis, we conducted a number of bias audits by observing differences in funding outcomes based on school type (magnet, charter, etc.), location (geographic NE/NW/SE/SW and political leaning), teacher prefix, and primary focus area. Some

of the audits conducted are found in the appendix, but we focused on primary focus area. Figure 6 highlights funding outcomes across different primary focus areas. The most common category is Literacy and Language, followed by Math & Science, and then Music & Arts. This distribution gives insight into which subjects dominate the platform and may help identify where support is most needed.

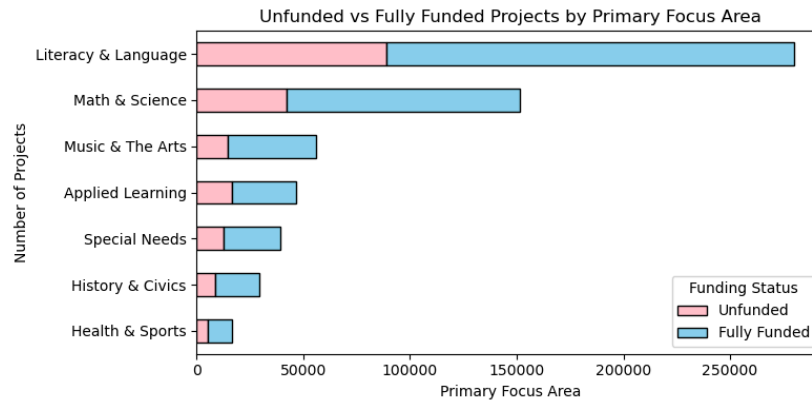


Figure 6. Unfunded vs fully funded projects by primary focus area.

Based on the data exploration and descriptive statistics, we have gathered sufficient information to proceed with addressing the problem at hand. The projects and outcomes tables will serve as the primary data sources for our analysis. These tables provide the necessary information about project submissions, their funding status, and the primary focus areas of each project, which we believe are key factors in building our predictive models. The insights gained from the initial exploration, including the funding imbalances and missing data, will guide us in data cleaning, feature selection, and model development to ensure accurate and meaningful results.

During data cleaning, we focused on a few key tasks: handling missing and duplicate values, dropping columns that weren't useful, checking for outliers, and making sure the data types were consistent. We checked for duplicate values both before and after merging the datasets to make sure everything was consistent and unique. Then we removed columns that didn't really add value to our model. For example, variables like `secondary_focus_subject` and `secondary_focus_area` weren't adding anything new since the `primary_focus_subject` and `primary_focus_area` already covered that information. We also dropped variables like `school_district`, `teacher_teach_for_america`, and `teacher_ny_teaching_fellow` because they didn't seem to be strong indicators for our prediction goals.

After narrowing down the dataset, we found that two variables had a lot of missing values. `fulfillment_labor_materials` had over 35,000 missing entries, and `school_metro` had even more, almost twice as many. A few other variables had smaller amounts of missing data, under 200 entries. To deal with this, we used different imputation methods depending on the type of data. For numerical columns, we filled in missing values with the mean. For categorical ones, like `primary_focus_area`, we used the mode (most common value). This helped us get a cleaner and more complete dataset, ready for the next steps in our project.

To deal with the outliers in the dataset, we started by creating the boxplots (Figure 7) for key numerical variables including `fulfillment_labor_materials`, `total_price_excluding_optional_support`,

total price including optional support, and students reached. These visualizations helped us identify the data points which were far beyond the main data range. At first, we considered using the IQR method to filter out the outliers, but it ended up excluding a large number of values that still seemed reasonable. As there would have been a lot of variations depending on the size of the school and the project expense, we ended up with a more targeted approach. Any of the negative values were removed, since costs and students cannot be negative, and then manually set the upper bounds for a few columns. These thresholds were as follows: \$5,000 for `'total_price_excluding_optional_support'`, \$6,000 for `'total_price_including_optional_support'`, and 1,000 for `'students_reached'`. They were chosen after carefully inspecting the boxplots. Most of the data was tightly clustered well below these values, and only a few extreme outliers (such as projects claiming costs in the millions or reaching hundreds of thousands of students) were far outside this range. These appeared to be either data entry errors or unrepresentative cases. By capping these values, we were able to clean the data without discarding too much of it. The dataset size only slightly decreased from 619,326 to 617,010 rows, but the overall quality and reliability of the data improved.

After cleaning the data, we preprocessed it by encoding categorical columns and converting booleans to 0s and 1s. We also created a new column for how many days had passed since a project was posted. This helped make everything numeric and ready for modeling, increasing our columns from 24 to 41. Furthermore, a correlation matrix was created to check for multicollinearity. As expected, some of the one-hot encoded variables showed negative correlations, but overall there were no strong correlations between features. I dropped `total_price_excluding_optional_support` since it was very similar to the "including" version.

For the feature selection, we initially tried identifying important features by looking at the logistic regression model's coefficients. However, this approach wasn't helpful, as most of the variables ended up having similar coefficient values and made it difficult to distinguish which ones were actually meaningful for prediction.

So instead, we sorted the features by their absolute coefficient values and added them to the model one by one to see if they made any difference. This method revealed that the first seven features, despite being ranked higher, didn't actually contribute to improving the model's accuracy or AUC. Their inclusion barely played a role in the prediction, almost like random guessing.

Based on that, we decided to drop those top seven and keep the rest. The remaining features were used to train a new logistic regression model, this time with class weights set to 'balanced' to deal with the imbalance in the target variable. This approach made the model simpler and more focused on the features that actually seemed to matter.

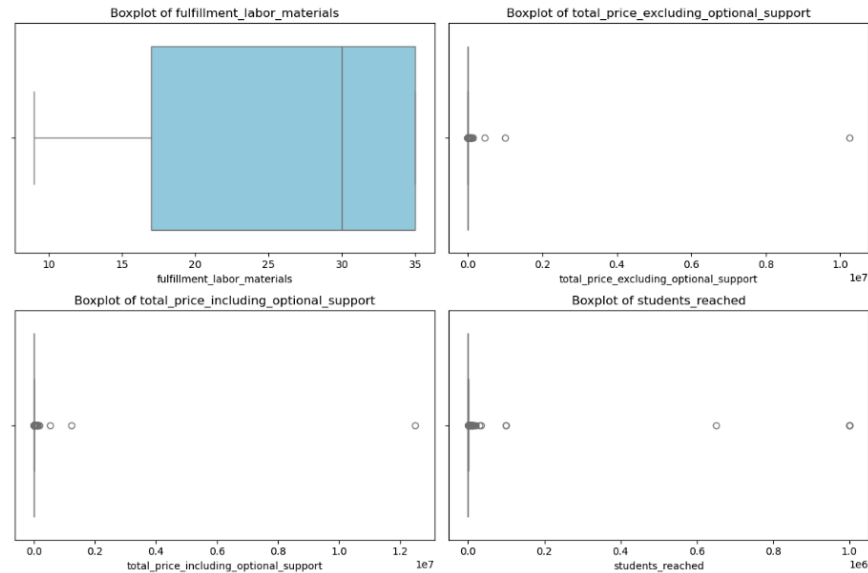


Figure 6. Boxplots for Numerical Variables

Analysis

For the project, we used supervised machine learning methods to perform a classification task: predicting whether a posted project will be fully funded before expiration. This predictive model will help DonorsChoose identify projects that are least likely to succeed without intervention, allowing a digital content expert to step in and improve those postings.

Given this, our outcome variable of interest was `fully_funded` (from `outcomes.csv`), which directly indicates whether a project was funded in full. Originally, we were also interested in using `is_exciting` (from `outcomes.csv`), which may act as an indicator for engaging projects that are likely to be funded. However, it appears that only a small percentage of projects were found exciting, and all of them were fully-funded.

To build the model, we evaluated several classification algorithms, including Logistic Regression, Naive Bayes, Gradient Boosting, and XGBoost, using 5-fold stratified cross-validation based on the ROC AUC metric. This robust evaluation on the training data indicated that XGBoost demonstrated the strongest baseline performance with a mean ROC AUC of 0.7663. Gradient Boosting followed with 0.7362, while Logistic Regression and Naive Bayes scored lower (0.6816 and 0.6313, respectively).

Based on these cross-validation results, we selected XGBoost for further optimization. We performed hyperparameter tuning using 3-fold stratified cross-validation Grid Search on the training data, optimizing again for ROC AUC. This process identified the optimal hyperparameters as: `colsample_bytree=0.9`, `gamma=0`, `learning_rate=0.1`, `max_depth=5`, `n_estimators=100`, and `subsample=0.7`. The best score achieved during the grid search cross-validation was 0.7559 ROC AUC.

Evaluating the final tuned XGBoost pipeline (incorporating necessary preprocessing like scaling within the pipeline structure used for CV/GS) on the held-out test set yielded an ROC AUC of

0.7572. At the default classification threshold of 0.5, this model achieved an accuracy of 0.75, with a precision of 0.75 and recall of 0.94 for predicting 'Funded' projects (class 1). However, its ability to identify 'Not Funded' projects (class 0) was weaker, with a precision of 0.67 and a recall (specificity) of only 0.29.

Since the primary objective involves accurately identifying projects likely to go unfunded (maximizing True Negatives/Specificity) while still capturing a reasonable number of at-risk projects, we analyzed the model's performance across different probability thresholds using the test set predictions. Optimizing for F1-score suggested a threshold of 0.47, which resulted in very high recall (0.96) but poor specificity (0.24). Optimizing instead for Youden's J statistic (Sensitivity + Specificity - 1) identified a more balanced optimal threshold of 0.68. At this threshold, the tuned XGBoost model achieved a Specificity (True Negative Rate) of 0.68 and a Recall (Sensitivity/True Positive Rate) of 0.68 on the test set. This 0.68 threshold offers a more practical trade-off for DonorsChoose's goal, enabling the identification of the target 10% of projects most likely to fail (by selecting those with predicted probabilities below a certain low percentile, informed by this threshold analysis) while better managing the digital content expert's workload compared to lower thresholds that generate more false positives.

We implemented our solution using Python, leveraging Pandas, Scikit-learn (for pipelines, scaling, cross-validation, grid search, and baseline models), XGBoost, and Matplotlib/Seaborn. The methodology involved the data cleaning steps established in the `ML_Final.ipynb` notebook, followed by applying a Scikit-learn pipeline incorporating StandardScaler (for relevant models like Logistic Regression) before the classifier. The final model utilized the features derived from the preprocessing, including project cost, student reach, school characteristics, and project details like days since posted.

Evaluation Metrics and Validation Techniques

Since the goal of this decision support system is to identify projects most in need of support—specifically, those least likely to be funded—our evaluation strategy focused on metrics reflecting the model's ability to correctly classify both funded and unfunded projects. While we initially used ROC AUC for model selection during cross-validation and hyperparameter tuning due to its robustness with imbalanced classes, our final threshold selection prioritized a balance between:

- **Specificity (True Negative Rate - Primary Focus):** Measures the proportion of truly unfunded projects (class 0) correctly identified as such. Maximizing this minimizes wasted intervention effort on projects that would succeed anyway.
- **Recall (Sensitivity/True Positive Rate - Secondary Focus):** Measures the proportion of truly funded projects (class 1) correctly identified. While not the primary goal, maintaining reasonable recall is important.
- **Precision:** Measures the proportion of projects predicted as funded that actually were funded.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a single measure of balance between them.
- **Youden's J Statistic: (Sensitivity + Specificity - 1),** used to find a threshold balancing TPR and TNR.

To ensure the model is reliable and generalizable, we employed the following validation techniques:

- **80/20 Train-Test Split:** We reserved 20% of the preprocessed dataset as a final, unseen test set to evaluate the tuned model's generalization performance.
- **5-Fold Stratified Cross-Validation:** Within the training data (80%), we used 5-fold stratified cross-validation to obtain robust baseline performance estimates for different model types (Logistic Regression, Naive Bayes, Gradient Boosting, XGBoost), guiding model selection.
- **3-Fold Stratified Cross-Validation (within Grid Search):** We used 3-fold stratified cross-validation internally within GridSearchCV to tune the hyperparameters of the selected best model (XGBoost) on the training data, optimizing for ROC AUC.
- **Threshold Analysis:** We analyzed the performance (Specificity, Recall, Precision, F1, Youden's J) of the final tuned model across various probability thresholds on the test set to select an optimal operating point (0.68 based on Youden's J) that best balanced the project objectives.

This multi-faceted validation approach helps ensure the selected model and threshold perform consistently and effectively on new, unseen project data and prioritize projects with the highest risk of going unfunded.

Results

The initial 5-fold cross-validation on the training data showed XGBoost (mean ROC AUC: 0.7663) significantly outperformed other baseline models like Logistic Regression (0.6816) and Naive Bayes (0.6313). Hyperparameter tuning via Grid Search yielded a best cross-validation ROC AUC of 0.7559 for XGBoost, indicating slight overfitting or that the default parameters were already near optimal for this metric on the CV splits.

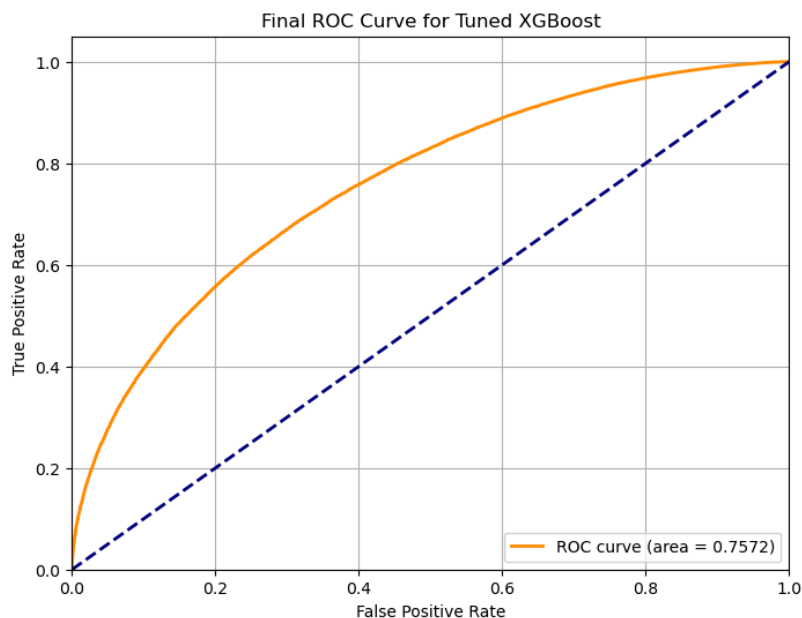


Figure 7. ROC Curve for Tuned XGBoost model

Evaluating the final tuned XGBoost model on the held-out test set with the optimized threshold of 0.68 (based on Youden's J) resulted in a Specificity of 0.68 and Recall of 0.68. This represents a significant improvement in identifying unfunded projects (Specificity) compared to using the default 0.5 threshold (Specificity: 0.29) or the F1-optimized threshold of 0.47 (Specificity: 0.24), albeit with a reduction in Recall (from 0.94 and 0.96, respectively). While maximizing

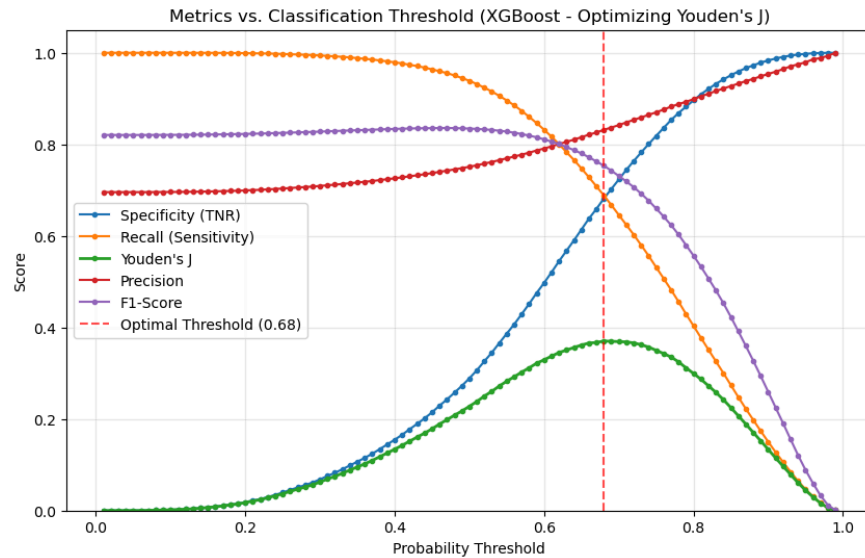


Figure 8. Metrics Performance across thresholds.

This chosen operating point better reflects the project goal of accurately flagging at-risk projects for the digital content expert. The final test set ROC AUC of 0.7572 confirms the model's overall discriminative ability remains strong.

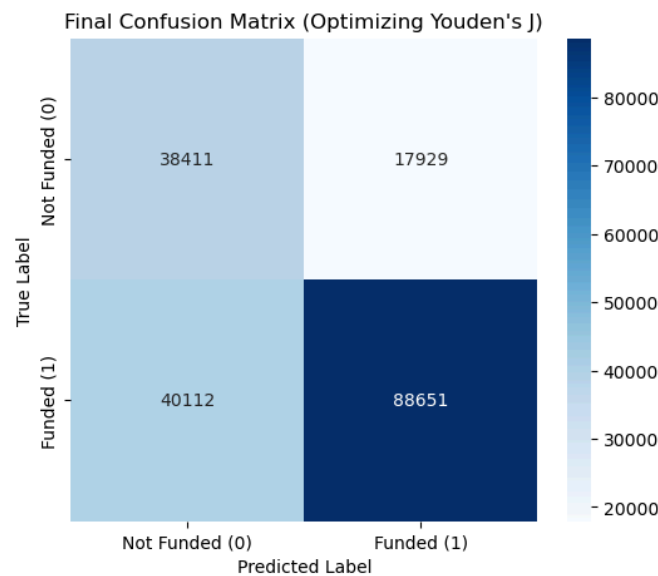


Figure 9. Confusion matrix at the chosen threshold.

Lastly, when comparing performance across groups where we suspected bias (poverty level and primary focus area), we found the following results:

Auditing by: poverty_level

Auditing by: primary_focus_area

Feature	Group	Group Size	Accuracy	Precision (Funded)	Recall (Funded)	Specificity (Not Funded)	F1 (Funded)	FPR	FNR
poverty_level	high poverty	48357	0.654	0.813	0.617	0.728	0.701	0.272	0.383
poverty_level	highest poverty	107283	0.720	0.842	0.757	0.623	0.797	0.377	0.243
poverty_level	low poverty	4612	0.593	0.806	0.516	0.749	0.629	0.251	0.484
poverty_level	moderate poverty	24851	0.622	0.812	0.533	0.780	0.643	0.220	0.467
primary_focus_area	Applied Learning	13942	0.679	0.805	0.655	0.721	0.722	0.279	0.345
primary_focus_area	Health & Sports	5052	0.690	0.851	0.676	0.722	0.753	0.278	0.324
primary_focus_area	History & Civics	8825	0.708	0.841	0.712	0.698	0.771	0.302	0.288
primary_focus_area	Literacy & Language	83799	0.672	0.830	0.655	0.710	0.732	0.290	0.345
primary_focus_area	Math & Science	45074	0.703	0.832	0.737	0.615	0.782	0.385	0.263
primary_focus_area	Music & The Arts	16814	0.723	0.846	0.767	0.596	0.805	0.404	0.233
primary_focus_area	Special Needs	11597	0.661	0.830	0.632	0.723	0.718	0.277	0.368

Figure 10. Bias audit table

Groups where specificity performance deteriorates are ‘music & the arts’ and ‘highest poverty’ level. This performance difference highlights the specific focus that these groups should receive when receiving targeted intervention, especially in ‘highest poverty’ groups that while achieving the highest accuracy across all the poverty levels is the least likely to detect projects that will go unfunded.

Policy Recommendations

Based on our analysis, we recommend implementing the tuned XGBoost model using a classification probability threshold of 0.68 to identify the bottom 10% of newly posted projects (those with the lowest predicted probabilities) for review by the digital content expert. This threshold, optimized using Youden's J statistic, provides a strong balance between correctly identifying truly unfunded projects (Specificity ≈ 0.68) and capturing funded ones (Recall ≈ 0.68), thereby focusing limited intervention resources effectively.

However, while the decision maker may worry about flagging several projects that would have been funded anyways, boosting the funding of projects at the decision boundary, which are vulnerable to receive no funding, could be a highly reasonable alternative to focus on projects most at risk that would require less intervention efforts to achieve the desired outcome change; therefore, we conclude that the obtained threshold and performance levels accomplish supporting the ultimate objectives effectively regardless of the ultimate approach the decision maker pursues to define the “10% most at risk of going unfunded projects.” Ultimately, we expect that the decision maker uses one of the two sections of the following table:

--- Top 15 Projects with LOWEST Funding Probability (Most At-Risk) ---

projectid	primary_focus_area	total_price_including_optional_support	poverty_level	Predicted Probability Funded	True Outcome (1=Funded)
8010778a12a2e860c6a003b6d16cbc88	Literacy & Language	\$3,158.23	moderate poverty	0.0463	0
bd12b4e95e7ae319fe425216afad8c81b	Literacy & Language	\$3,524.26	highest poverty	0.0508	0
6b4bcc7cfbcd8256e748f44e26765fc5	Literacy & Language	\$1,811.54	high poverty	0.0531	0
3f93d2b47e7aee8f98a3f967e1c8be9	Literacy & Language	\$4,603.99	highest poverty	0.0552	0
9729bbda888273f603906b173ede7457	Literacy & Language	\$529.74	moderate poverty	0.0601	0
974608ea2915c42bbfc6f68aa8ceb531	Math & Science	\$823.57	high poverty	0.0622	0
2dfad30a8f72daa1b931a1c37b3c929a	Literacy & Language	\$940.16	highest poverty	0.0649	0
43404d64fe4a1a527221e77c9428c493	History & Civics	\$840.29	high poverty	0.0649	0
9bab1577f253f967711e463d1f9ede31	Math & Science	\$1,848.54	high poverty	0.0650	0
ef97591f0407d5561360814cb422c7f4	Literacy & Language	\$979.20	high poverty	0.0650	0
38cf73f8e53d8fe63652b93097b28fc8	Literacy & Language	\$1,457.97	highest poverty	0.0655	0
27b65473c9348b5d3e9389a12e731fff	History & Civics	\$946.80	highest poverty	0.0663	0
d62cba1e8e0343c82fd6528a1f9ae31f	Literacy & Language	\$431.59	high poverty	0.0667	0
fa984400b3514a29f5ca12d20738451a	Applied Learning	\$744.53	moderate poverty	0.0672	0
b673a5ed310ad7d3e4630940fae9e1a	Literacy & Language	\$987.34	high poverty	0.0673	0

--- Top 15 Projects JUST BELOW Optimal Threshold (0.68) ---

projectid	primary_focus_area	total_price_including_optional_support	poverty_level	Predicted Probability Funded	True Outcome (1=Funded)
b60c9df526b06cda6a94143e9804543c	Literacy & Language	\$430.27	high poverty	0.6800	0
ca519fb064a5f0c8015d068a8e68067b	Literacy & Language	\$571.46	highest poverty	0.6800	1
0cff87e279b6c80b3b905896a41d0c0e	Literacy & Language	\$959.96	high poverty	0.6800	1
1b8137f463a2e64bfd34783a596714ba	Literacy & Language	\$940.47	high poverty	0.6800	1
7d308c1325b5c545c380026d3bccbda7	Literacy & Language	\$438.22	highest poverty	0.6800	0
f1c17a6d5f0ed85efa1ae78ef541507e	Literacy & Language	\$276.80	highest poverty	0.6800	1
d0e28ccb28c8dfd8fb27f080fac1a5cb	Math & Science	\$586.91	high poverty	0.6800	0
6647d01226b096e189cb3adaa0c8f60e	Literacy & Language	\$654.89	high poverty	0.6800	1
3591ab4b357137d14a21db0b555bf15a	Math & Science	\$1,040.56	low poverty	0.6800	1
ab4aa72d47236dfe7b1620e0c84ae3	Literacy & Language	\$547.56	highest poverty	0.6800	0
35006033685325c4a790e2b7631ad08c	Math & Science	\$451.72	highest poverty	0.6800	1
fbfb2b4ae64d1edbb6d65f0f8e9db39b	Math & Science	\$536.41	highest poverty	0.6800	1
4f1887162ba3f4cdcfaef44271bee022	Special Needs	\$1,081.71	highest poverty	0.6800	1
ed0f3ec0f13a3880a3c7fb35c5665ace	Literacy & Language	\$463.56	highest poverty	0.6800	1
ac533b3aaebc7584560029fb934932d7	Literacy & Language	\$268.07	highest poverty	0.6800	0

Figure 11. Table of projects with lowest and right-below-the-cutoff probabilities of funding

To ensure sustained success and fairness, policy adjustments should include:

- **Periodic Model Recalibration:** Retrain the model quarterly or semi-annually using updated data and appropriate validation techniques (like temporal cross-validation) to adapt to potential shifts in project characteristics or donor behavior.
- **Performance Monitoring:** Continuously track the model's real-world performance, particularly its Specificity and Recall at the chosen 0.68 threshold, to ensure it consistently meets operational goals for identifying at-risk projects.
- **Intervention Impact Analysis:** Implement A/B testing where feasible to measure the actual funding lift achieved by the digital content expert's interventions on flagged projects. This feedback loop is crucial for refining the intervention strategy and demonstrating value.
- **Fairness Audits:** Continuously monitor key fairness metrics (e.g., true negative rates and false negative rates across different school poverty levels, geographic regions, or project focus areas) to ensure the intervention strategy equitably benefits under-resourced communities and adjust tactics as needed.

Limitations and Future Work

Key challenges encountered during this analysis include potentially incomplete outcome data and the inherent class imbalance between funded and unfunded projects. The discrepancy between the number of projects submitted and those with final outcomes suggests ongoing funding cycles or missing data, requiring careful data handling. Missing values in certain features also necessitated imputation, which could influence model predictions. Furthermore, the dynamic nature of crowdfunding platforms means that donor behavior and project trends may evolve, requiring periodic model updates to maintain predictive accuracy. Specific limitations of the current model and analysis include the inherent threshold trade-off between correctly identifying funded versus unfunded projects, the exclusion of potentially valuable donation and resource data in this iteration, and the focus on a standard feature set without extensive feature engineering. In future instances, working with decision makers would allow us to create differentiated thresholds that reduce specificity biases across groups susceptible to these phenomena.

Future work should prioritize integrating this predictive model into a comprehensive decision support system equipped with database management tools, ad hoc analysis capabilities, and a user-friendly interface for DonorsChoose staff, particularly the digital content expert. A crucial enhancement to this support system would be the inclusion of intervention analysis features, enabling systematic tracking and evaluation of the funding impact resulting from editorial changes made to flagged projects. Further research could explore alternative modeling techniques specifically designed for imbalanced datasets, investigate the value of incorporating features derived from project text descriptions using natural language processing, and revisit more complex feature engineering to potentially improve prediction accuracy, fairness, and the overall effectiveness of the intervention strategy.

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Supplementary Figures and Tables

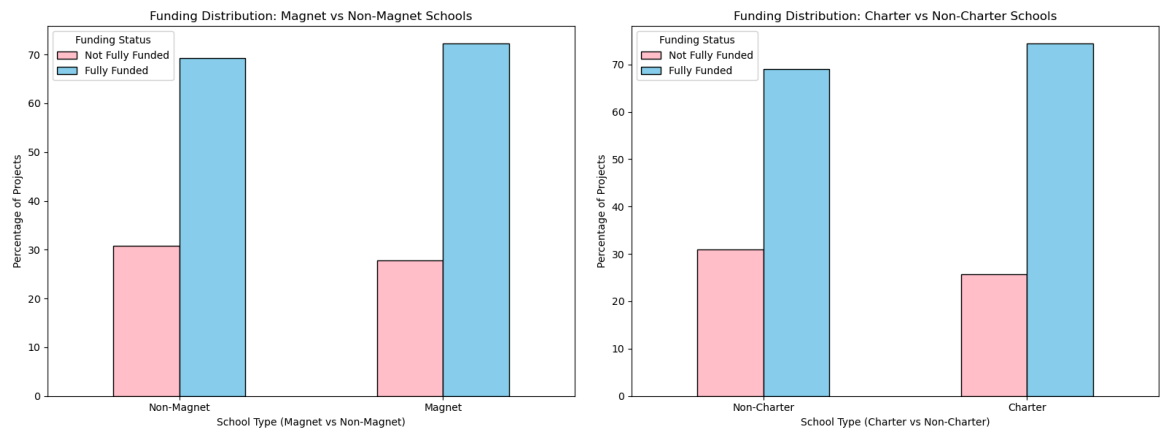
Supplementary Table 1. Tables and attributes in the dataset.

Table	Columns
outcomes	['projectid', 'is_exciting', 'at_least_1_teacher_referred_donor', 'fully_funded', 'at_least_1_green_donation', 'great_chat', 'three_or_more_non_teacher_referred_donors', 'one_non_teacher_referred_donor_giving_100_plus', 'donation_from_thoughtful_donor', 'great_messages_proportion', 'teacher_referred_count', 'non_teacher_referred_count']
projects	['projectid', 'teacher_acctid', 'schoolid', 'school_ncesid', 'school_latitude', 'school_longitude', 'school_city', 'school_state', 'school_zip', 'school_metro', 'school_district', 'school_county', 'school_charter', 'school_magnet', 'school_year_round', 'school_nlns', 'school_kipp', 'school_charter_ready_promise', 'teacher_prefix', 'teacher_teach_for_america', 'teacher_ny_teaching_fellow', 'primary_focus_subject', 'primary_focus_area', 'secondary_focus_subject', 'secondary_focus_area', 'resource_type', 'poverty_level', 'grade_level', 'fulfillment_labor_materials', 'total_price_excluding_optional_support', 'total_price_including_optional_support', 'students_reached', 'eligible_double_your_impact_match', 'eligible_almost_home_match', 'date_posted']

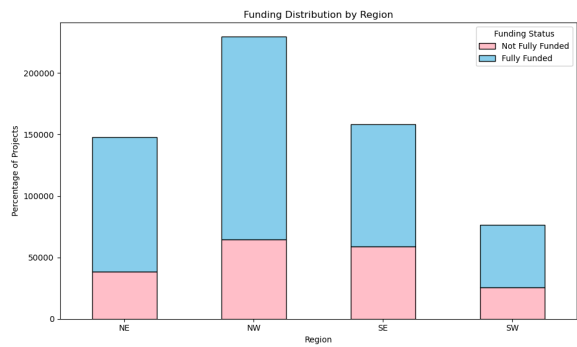
Missing Values:		teacher_prefix	4
projectid	0	teacher_teach_for_america	0
teacher_acctid	0	teacher_ny_teaching_fellow	0
schoolid	0	primary_focus_subject	39
school_ncesid	42736	primary_focus_area	39
school_latitude	0	secondary_focus_subject	207893
school_longitude	0	secondary_focus_area	207893
school_city	0	resource_type	45
school_state	0	poverty_level	0
school_zip	4	grade_level	9
school_metro	81908	fulfillment_labor_materials	35082
school_district	948	total_price_excluding_optional_support	0
school_county	17	total_price_including_optional_support	0
school_charter	0	students_reached	146
school_magnet	0	eligible_double_your_impact_match	0
school_year_round	0	eligible_almost_home_match	0
school_nlns	0	date_posted	0
school_kipp	0	dtype: int64	

Supplementary Figure 1. Missing values for projects table.

Further Exploratory Data Analysis



Geographic Distribution



Political Distribution

